

Aws Abdulfattah

Forward Deployed Engineer · Payments Infrastructure, AI, Field Delivery

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Senior engineer who ships inside other people's constraints — regulators, legacy stacks, unreliable networks, and code someone else wrote.

At **Qi Card**, Iraq's largest fintech, I built **Tasdid**, the pay-with-Qi payment service that third-party apps integrate, and **Tasdid+**, the bill-payment app on top of it (**1M+ users, ~500K transactions/month**) — taking flows through Central Bank of Iraq review to get them live. At **Cardyiq** I took over a failing agency-built product days after launch and rebuilt it in production into a sports network across **150+ venues in 6 cities**. Five startups have since hired me to integrate the payment rail I built.

Twelve years building systems people depend on; the last five in software. Before that, nine as an architect — including managing post-conflict housing reconstruction for UNOPS in Anbar. I learned to program to automate my own drawings, and never went back. I am comfortable in the field, in front of customers, and in someone else's codebase.

SKILLS

Working in the field — Regulated environments, customer-facing delivery, other people's codebases, constrained networks and low-end devices.

Central Bank of Iraq compliance Third-party integration Production rebuilds Incident response Customer discovery

Payments — Built the rail and the apps on top of it; now integrate it for other companies.

Card tokenization Idempotent endpoints Reconciliation & settlement Webhooks Redirect/hosted flows

Languages

TypeScript JavaScript Python SQL

Frontend & Mobile

React Svelte / SvelteKit Next.js React Native Tailwind

Backend & Data

Node.js Express Bun Prisma REST design

Infrastructure

Linux / VPS Docker GitHub Actions Dokploy DNS

Distributed systems

BullMQ RabbitMQ Idempotency Retries & DLQ Rate limiting

AI

OpenAI API Gemini Prompt engineering Coding agents MCP

EXPERIENCE

Cardyiq — Founding Engineer & Technical Lead

September 2025 — Present · concurrent, equity
Baghdad, Iraq

- Joined days after an agency-built launch failed. The app was **unusable without a VPN** (DNS/CDN misconfiguration), had no caching layer, no CI/CD, an English-only dashboard in an Arabic-speaking market, and cash-only payments. Rebuilt backend, dashboard and mobile into a modular system — **live, without taking the product down**.
- Diagnosed that the constraint was commercial, not technical: replaced a confusing entry-pass model with a points system, which unlocked expansion beyond gyms into pools, padel and equestrian venues — now **150+ venues across 6 Iraqi cities**.
- Integrated card and wallet payments (Wayl, and Tasdid pay-by-Qi), eliminating manual cash handling. Wayl cited Cardyiq's integration as the **best they had seen, by authorization success rate**. Qi-wallet payments now carry roughly half of all transactions.
- 1,200 active subscribers, 10K+ installs, 12K+ venue check-ins, ~\$53K (70M IQD) processed** in 7 months.
- Built CI/CD from nothing (Docker, GitHub Actions, Dokploy) and migrated hosting for regional latency. **99.9% uptime over 8 months**; recovered a data-loss incident with **zero records lost** by promoting a read replica.
- Hired and lead 5 engineers across backend, frontend, mobile, UI/UX and QA. 30 mobile releases in 6 months.
- Stack: Flutter, React, TypeScript, MySQL, Docker, GitHub Actions, Dokploy, Linode, Laravel (inherited).

Qi Card — Senior Software Engineer — Billing

November 2024 — Present
Baghdad, Iraq

- Billing team at Iraq's largest fintech. I build inside **SuperQi**, Qi Card's super-app, on an Alipay-based mini-app runtime. Promoted to Senior in March 2026.
- Built **Tasdid**, the pay-with-Qi payment service that **third-party applications integrate** to accept Qi Card payments — and **Tasdid+**, the consumer bill-payment mini-app on top of it: **1M+ users, ~500K transactions/month**.
- Worked directly against **Central Bank of Iraq** requirements: designed and defended a peer-to-peer invoicing flow through regulatory review, and implemented card-payment support in Tasdid+ to satisfy CBI's anti-monopoly rule that any government bill must be payable by any card.
- Hardened the P2P flow after internal security review — OTP transaction verification, payer identity removed at initiation. Every mini-app I ship passes Qi's internal security review: encrypted API transport, strict auth layers, idempotent payment endpoints.

- Built **Jabi+** for merchants who run their business from a phone, not a laptop (transactions, statements, payroll, supplier payments, Excel export), and **e-Psule**, trilingual Kurdish/Arabic/English billing for the Kurdistan Region across providers with wildly different response times.
- Chose **Svelte** deliberately: the mini-app runtime is a constrained SPA host, and users are on low-end Android over slow networks. Smallest bundle, fastest cold start.
- Mentor junior engineers and teach students through the Qi Card–AUIB programme.

Independent — Payments Integration Consultant

2025 — Present
Iraq · Remote

- Retained by 5 startups to integrate Iraq's payment rails — **Tasdid (pay-by-Qi)** and **Wayl** — into mobile and web products. Clients include Orivo, 3aljni and Skincare by Self.
- I built the Tasdid service itself, so integrations land first-time-right: correct redirect/webview handling, idempotency, webhook reconciliation, and checkout flows that don't lose users mid-payment.

HAWSR — Founder

July 2020 — August 2023
Baghdad, Iraq

- Turned my own master's thesis into a company. Built a 3D property-visualisation web platform (Three.js, React Three Fiber, React, Tailwind) to **~\$60K in revenue**.
- Automated the Autodesk Revit → web delivery pipeline with Dynamo and Python, removing manual model preparation.
- R&D: a pipeline converting 360° captures (Insta360 SDK, Python) into interactive **3D Gaussian Splatting** scenes for real-estate and photography use.
- Wound down. The market was too small, funding ran out, and splatting quality at that time did not scale. **1st Prize, The Station Foundation Incubator; Most Innovative Startup Founder, CS50x Baghdad**. Presented HAWSR on stage to an audience of 1,000+.

UNOPS — BIM Consultant — Anbar Reconstruction

August 2022 — August 2023
Anbar, Iraq

- Managed reconstruction of housing for families displaced by the ISIS conflict in Anbar — contractors, budgets, delivery schedules and BIM coordination under field conditions. Wrote code nights and weekends throughout, and joined my first startup engineering role before the contract ended.

SELECTED WORK

Tasdid & Tasdid+ — National Bill Payment (Qi Card)

2024 — Present

Tasdid is the pay-with-Qi payment service: third-party applications integrate it to accept Qi Card payments without handling card data. **Tasdid+** is the consumer mini-app on top — pay government and private bills across many providers through one interface. **1M+ users, ~500K transactions/month**. SuperQi (Alipay-based) APIs for authentication, bill enquiry, invoice display, wallet and card-tokenized payments. Svelte architecture separating UI from state; caching and progressive loading to absorb providers with very different response times. Designed against Central Bank of Iraq requirements.

Jabi+ — Merchant Transactions Mini-app (Qi Card)

2025 — Present

A merchant mini-app for the businesses that **don't have a laptop** — 1–10 people running their finances from a phone. Incoming transactions, statements, payroll to employees, payments to business services.

Filtering, sorting and pagination over large transaction datasets, plus Excel export for reconciliation and reporting. Mobile-first, built for operational use by real business accounts.

NotifyX — Open-source Notification Infrastructure · github.com/awsdang/notifyx

2025 — Present · Open source

Multi-channel notification platform: FCM, APNS, HMS and Web Push behind one API, with plug-and-play SDKs for web, Flutter and React Native.

Backend API, worker and scheduler in **Node.js, BullMQ, Redis and PostgreSQL (Prisma)** — asynchronous job processing, scheduling, retries and queue orchestration. Load-tested with k6 at **~150 messages/sec per worker**. React admin dashboard for delivery operations.

EDUCATION

Turing College —Certificate, AI Engineering

October 2025 — 2026

Completed. Capstone: Maaahr, an AI interview-practice platform, later extended to live voice.

University of Technology —B.Sc., Architectural Engineering

October 2012 — July 2017

University of Ljubljana —Erasmus Exchange, MSc Information Management (coursework & thesis)

October 2020 — October 2021

Coursework and thesis completed remotely; degree not conferred (visa denied during COVID-19). The thesis was commercialized as **HAWSR**.

AWARDS, CERTIFICATIONS, COMMUNITY & LANGUAGES

1st Prize — Incubator Program The Station Foundation · 2022	AI Engineering Turing College · 2026	The Station Hackathon 2023 — Present	Arabic Native
Most Innovative Startup Founder in Baghdad CS50x Baghdad · 2022	Machine Learning for Inverse Graphics & AI MIT · March 2024	Startup Weekend Baghdad 2017 — 2023	English Professional — IELTS 7.5–8
	Python — AI & Machine Learning 10K Arab AI Developers Initiative · January 2024	Toastmasters International 2020 — 2021	Dutch Beginner
	Three.js Journey Bruno Simon · July 2022	Qi Card × AUIB 2025 — Present	